

Achieving Ultrahigh Energy Densities of Supercapacitors with Porous Titanium Carbide/Boron-Doped Diamond Composite Electrodes

Jing Xu, Nianjun Yang,* Steffen Heuser, Siyu Yu, Anna Schulte, Holger Schönherr, and Xin Jiang*

The energy densities of most supercapacitors (SCs) are low, hindering their practical applications. To construct SCs with ultrahigh energy densities, a porous titanium carbide (TiC)/boron-doped diamond (BDD) composite electrode is synthesized on a titanium plate that is pretreated using a plasma electrolytic oxidation (PEO) technique. The porous and nanometer-thick TiO_2 layer formed during PEO process prevents the formation of brittle titanium hydride and enhances the BDD growth during chemical vapor deposition processes. Meanwhile, the in situ conversion of TiO_2 into TiC is achieved. Combination of this capacitor electrode with soluble redox electrolytes leads to the fabrication of high-performance SCs in both aqueous and organic solutions. In 0.05 M $\text{Fe}(\text{CN})_6^{3-/4-} + 1 \text{ M Na}_2\text{SO}_4$ aqueous solution, the capacitance is as high as 46.3 mF cm^{-2} at a current density of 1 mA cm^{-2} ; this capacitance remains 92% of its initial value even after 10 000 charge/discharge cycles; the energy density is up to 47.4 Wh kg^{-1} at a power density of 2236 W kg^{-1} . The performance of constructed SCs is superior to most available SCs and some electrochemical energy storage devices like batteries. Such a porous capacitor electrode is thus promising for the construction of high-performance SCs for practical applications.

1. Introduction

Supercapacitors (SC) and batteries are the most promising electrochemical energy storage (EES) devices.^[1,2] SCs, also named as electrochemical capacitors or ultracapacitors, store electrical energy in the form of an electrical double layer capacitor (EDLC) or a pseudocapacitor (PC). The former is

based on charge/ion accumulation at the electrode/electrolyte interface, the latter is based on electron transfer from the Faradaic reactions of redox species coated on the surface of the capacitor electrode or dissolved in the solution.^[2–7] Compared to batteries, SCs show higher power densities, shorter charging/discharging times, and longer lifetime. Hence, SCs can complement or even replace the batteries in some industrial applications.^[2–7] However, the major challenge of currently existed SCs is their low energy densities, typically in the range of $3\text{--}5 \text{ Wh kg}^{-1}$.^[2–7] To realize their wider and more industrial applications, SCs with high power and energy densities, also called battery-like SCs, supercapacitors, or supercapacities, are thus highly demanded.

To produce such SCs, suitable electrolytes are required. For example, to enhance the energy densities of SCs, one effective strategy recently employed is to introduce soluble redox-active species (e.g., ferro-

cene, $\text{Fe}(\text{CN})_6^{3-/4-}$) into the solutions or to coat redox-active species (e.g., conducting polymers, metal oxides) onto the capacitor electrodes.^[2–7] The PCs constructed using porous capacitor electrodes (e.g., activated carbon) have exhibited much higher capacitances than EDLCs. Taking the capacitance of a PC using $\text{Fe}(\text{CN})_6^{3-/4-}$ redox couples as an example, it is about four-order higher than that of a diamond EDLC.^[3] This is because this pseudocapacitance originates from fast Faradaic reactions of $\text{Fe}(\text{CN})_6^{3-/4-}$ redox couple. Eventually such a PC provides much higher energy densities than a EDLC. Another key to fabricate battery-like SCs, supercapacitors, or supercapacities is to select a perfect capacitor electrode. As the most prospective capacitor electrodes, sp^2 carbon based materials (e.g., activated carbon, carbon nanofibers, carbon nanotubes, graphene, and graphene related materials)^[2,4,5] have been extremely utilized. Recently, sp^3 carbon based materials such as synthetic conductive diamond (e.g., boron-doped diamond, BDD) films and their micro-/nanostructures have been also employed as the capacitor electrodes.^[6–9] These flat diamond films are often grown on Si substrates by means of chemical vapor deposition (CVD) technique, while the approaches of top-down etching, bottom-up overgrowth, and template-free etching are utilized to produce

J. Xu, Dr. N. Yang, S. Heuser, Dr. S. Yu, Prof. X. Jiang
Institute of Materials Engineering
University of Siegen
57076 Siegen, Germany
E-mail: nianjun.yang@uni-siegen.de; xin.jiang@uni-siegen.de

A. Schulte, Prof. H. Schönherr
Physical Chemistry I & Research Center of Micro and Nanochemistry
and Engineering (Cμ)
Department of Chemistry and Biology
University of Siegen
57076 Siegen, Germany

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/aenm.201803623>.

DOI: 10.1002/aenm.201803623

diamond micro-/nanostructures.^[6] Compared with those sp² carbon based materials, diamond capacitor electrodes exhibit more advantages.^[6] For example, they own large electrochemical potential windows in various media and long-term stability under extremely harsh conditions. Different from most sp² carbon based materials that actually exist in the form of powders or particles, diamond capacitor electrodes require neither current collectors nor organic binders. The application of diamond micro-/nanostructures (e.g., porous diamond,^[7] diamond nanowires,^[8] honeycomb diamond,^[9] etc.) as the capacitor electrodes enhances efficiently the capacitances (in the range of 3–7 $\mu\text{F cm}^{-2}$)^[6] of a flat diamond capacitor electrode, determined by the enlarged surface areas. However, to grow conductive diamond films, the brittle silicon (Si) substrates are often used. To form diamond micro-/nanostructures, the masks are needed for both top-down etching and bottom-up overgrowth approaches. Nevertheless, the preparation of these masks is complicated, time-consuming, and expensive in most cases. When a top-down etching process is applied, part of valuable diamond is lost. Indeed, these drawbacks of diamond capacitor electrodes hinder greatly fast development and further practical applications of diamond SCs.

We are thus interested in designing and synthesizing various kinds of diamond capacitor electrodes.^[6] They must be free of binders or current-collectors and feature large surface areas. For new types of diamond capacitors, we attempt to replace the brittle and metal-like Si substrate with a metal plate such as titanium (Ti) since it is lighter and cheaper than others (e.g., niobium, molybdenum, tantalum, etc.). Therefore, as-grown diamond capacitors will be easier to handle due to improved mechanical properties. Their conductivities are expected to be better than those grown on Si substrates. However, during CVD growth of diamond on a Ti substrate the formation of brittle titanium hydrides (e.g., TiH₂, TiH_{1.971}) occurs, resulting from the reaction of Ti with hydrogen in a CVD chamber.^[10] To avoid the formation of brittle titanium hydrides, a layer of titanium oxides (e.g., TiO₂) formed on the surface of a Ti substrate is helpful since it prevents the reaction of hydrogen with the Ti substrate. For example, CVD overgrowth of micrometer-long and vertically aligned TiO₂ nanotubes that formed using the anodic oxidation approach has been conducted, leading to the production of different BDD micro-/nanostructures.^[11] Similar bottom-up approach has been applied to overgrow the template based on carbon nanotubes with a thin BDD layer.^[12] Electrochemical properties (mainly capacitive behavior) of these BDD structures have been investigated. Although neither SC devices (including EDLCs and PCs) nor the SC performance (e.g., capacitance retention, power and energy densities, SC demonstrators, etc.) was provided, these BDD micro-/nanostructures were believed to be promising capacitor electrodes for the SC construction and further for other EES applications.^[11,12]

Herein, we report about the CVD synthesis of a porous TiC/BDD composite film on a Ti substrate, rather than micro-/nanostructures only consisting of BDD.^[11,12] The pretreatment of a Ti substrate using an anodic oxidation process (e.g., at a voltage of 60 V, in 0.1 M NH₄F solutions for 1 h)^[11] is replaced by a plasma electrolytic oxidation (PEO) process (e.g., at a high voltage of 280 V, in 0.066 M Na₃PO₄ aqueous solution, and for a short time of 15 min). The generated TiO₂ layer is porous

and only nanometer thick, in comparison to micrometer-long and vertically aligned TiO₂ nanotubes produced using an anodic oxidation method.^[11] During CVD growth process, the BDD phase is grown on diamond nanoparticles seeded on the PEO-treated Ti substrate. Meanwhile, the TiC phase is in situ formed by thermal reduction of TiO₂ with hydrocarbon ions and radicals generated during CVD growth of BDD. Since titanium carbides are highly conductive and have been proved to be promising electrode materials for the construction of EES devices,^[13] such porous, binder- and current-collector-free TiC/BDD composite films are explored as the capacitor electrodes to fabricate high-performance and battery-like SCs. The applied solutions cover both aqueous and organic media in the absence and presence of soluble redox electrolytes.^[14] The performance of constructed SCs (e.g., their capacitances, capacitance retention, power, and energy densities) is studied in detail. A SC prototype that supplies to light a LED is demonstrated. To our best knowledge, the combination of such a porous TiC/BDD composite electrode with soluble redox electrolytes to construct SCs with big and stable capacitances as well as high power and energy densities has not been reported yet. More importantly, achieved energy and power densities of these SCs are found to be much higher than most currently available SCs and some EES devices such as batteries. This study thus brings in new concepts to develop high-performance and battery-like SCs, especially those with ultrahigh energy densities.

2. Results and Discussion

2.1. Growth of Porous TiC/BDD Composite Films

The morphology of a TiO₂ layer formed on a Ti substrate during the PEO treatment, was firstly investigated using SEM. **Figure 1a,b** show its SEM images in the top and side view, respectively. Clearly, a volcano-like oxide layer with a relative smooth surface is formed on the Ti plate after the PEO treatment (Figure 1a). The formation of this porous surface is caused by the release of gas bubbles during the PEO treatment.^[15] Among 197 pores counted from Figure 1a, their pore sizes are varied from 0.01 to 2.2 μm . The pores with the diameters of 0.2–0.8 μm have the distribution ratio of about 55% (Figure S1, Supporting Information). The calculated average porosity (namely density of pores) is about $6 \pm 0.1\%$. As measured from its cross-section SEM image (Figure 1b), the thickness of this TiO₂ layer is about 0.8 μm .

In order to clarify the influence of a CVD process on the morphology of synthesized composite films, their microstructures (mainly pore sizes and pore-size distribution) after CVD growth of BDD for different deposition times were examined. Figure 1c shows the example SEM image of a TiC/BDD composite film grown with a deposition time of 30 min. One can see that the BDD nanoparticles or nanocrystals (with an average diameter of 200 nm) partially cover the surface. The calculated coverage ratio is about 75%. Interestingly, some BDD nanoparticles or nanocrystals are found deeply inside the micropores, probably due to favorable seeding/nucleation of diamond nanoparticles at the edge or dimples around these pores. When the growth time is longer than 60 min, the surface is almost fully covered

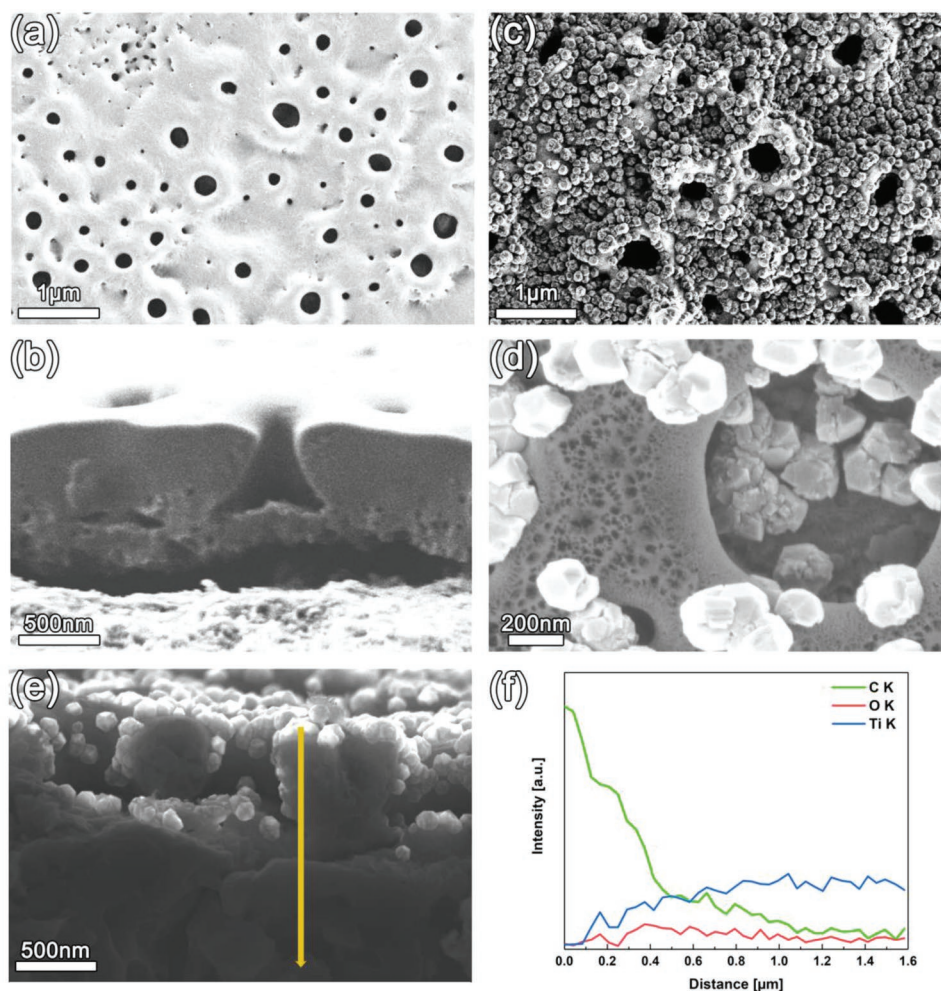


Figure 1. SEM images of a,b) a porous TiO₂ layer formed during the PEO treatment of a Ti substrate and c–e) a porous TiC/BDD composite film in a,c) top and b,e) side views. Image (d) is a magnified SEM image of (c). Plot (f) represents the EDX line profile along the arrow in the SEM image (e).

with a nanocrystalline BDD film where only few pores are visible (Figure S2, Supporting Information). In contrast, to fully cover a Ti substrate (without PEO treatment) with a same BDD layer, a growth time of 8 h was required under identified growth conditions.^[16] Moreover, without seeding of diamond nanoparticles on a PEO-treated Ti substrate, only a TiC film was grown but no growth of BDD nanocrystals/nanoparticles was found (Figure S3b, Supporting Information). Consequently, the porous TiO₂ layer formed during the PEO treatment is helpful to nucleate or seed diamond nanoparticles, eventually leading to improved growth rates of BDD nanoparticles/nanocrystals. In addition, the surface of a TiC/BDD composite film is much rougher than that of a TiO₂ layer formed on a Ti substrate during the PEO treatment. Interestingly, the PEO-treated Ti substrate after being etched by hydrogen plasma features similar structure and surface roughness (Figure S3a, Supporting Information) with this TiC/BDD composite film. Therefore, the increased surface microroughness of a TiC/BDD composite film is due to the hydrogen plasma etching. Furthermore, the microporous structure of a PEO-treated Ti substrate does not vary much after a CVD growth of BDD for 30 min. Namely a TiC/BDD composite film is porous and has the same surface

structure as the porous TiO₂ layer formed during this PEO process. Besides these micropores on the surface of a TiC/BDD composite film, some nanopores are also visible (Figure 1d). These tiny nanopores are probably generated by an additional carbonization process and/or by hydrogen plasma etching during a CVD process.

The cross-sectional SEM image of this porous TiC/BDD composite film (Figure 1e) reveals that it consists of two layers: a porous top layer and a compact thin inner layer. An interlayer is seen between BDD and a Ti substrate. The estimated depth (namely thickness) of such an interlayer is about 750 nm, slightly thinner than that (0.8 μm) of the porous TiO₂ layer formed during a PEO process. The thickness variation is believed to originate from hydrogen plasma etching of TiO₂ during CVD process.^[11] The EDX line profile was further employed to analyze the element distribution across this composite film. Figure 1f reveals that the content of carbon element is reduced with an increase of the depth, while the Ti content distribution shows an opposite trend. Very weak signal of oxygen element is detected along the cross-sectional film, an indication of full conversion of TiO₂ into TiC during CVD growth of BDD.

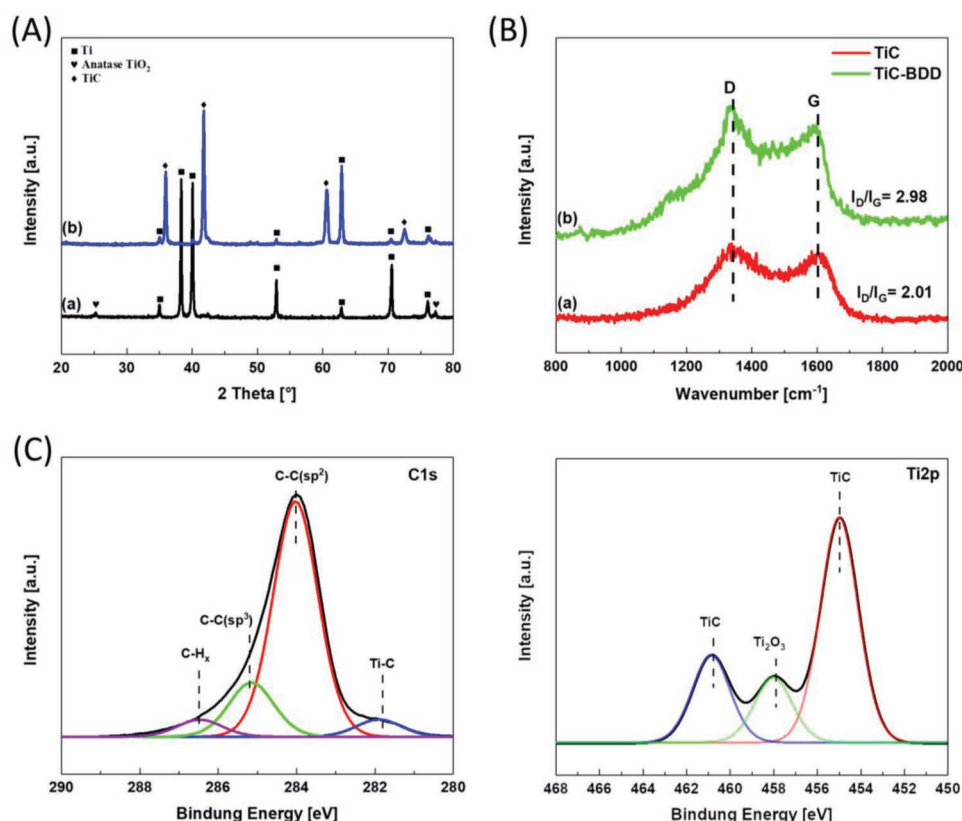


Figure 2. A) XRD patterns of a) a TiO₂ layer formed on a Ti substrate during the PEO treatment and b) a porous TiC/BDD composite film; B) Raman spectra of a) a TiC film and b) a porous TiC-BDD composite film; C) XPS spectra of C 1s (left) and Ti 2p (right) for a porous TiC/BDD composite film.

X-ray diffraction (XRD) was then used to analyze the crystalline structures and phase compositions of a TiO₂ layer (Figure 1a) and a porous TiC/BDD composite film (Figure 1c). The diffraction patterns of a TiO₂ layer (Figure 2A-a) reveal only relatively weak signal of anatase TiO₂. Namely, only a fraction of TiO₂ in such a porous layer is crystallized. Most TiO₂ formed during the PEO treatment is amorphous.^[17] For a porous TiC/BDD composite film, its XRD patterns (Figure 2A-b) are different. The diffraction patterns at the 2θ of 36.1°, 41.7°, 60.7°, and 72.6° are in good agreement with those for TiC materials (ICDD Card: 00-001-1222), confirming the presence of TiC phase in the composite film. Moreover, the diffraction patterns of anatase TiO₂ phase are not visible. Therefore, the formation of TiC originates from a carbonization process, namely thermal conversion of TiO₂ into TiC during the CVD growth of BDD. Such a conversion is actually quite reliable since the CVD deposition of BDD is carried out at a relatively high temperature (>700 °C), whereas the conversion of TiO₂ into TiC occurs around 200–450 °C.^[18] Furthermore, neither the diffraction patterns of TiH₂ (at 2θ of 59.64°, JFPDS Card No. 9-371) nor the peaks of TiH_{1.971} (at 2θ of 58.80°, JFPDS Card No. 7-370) are noticed. However, titanium hydrides (e.g., TiH₂ phase) are clearly detected once a Ti substrate (without PEO pretreatment) is exposed to hydrogen plasma, confirmed from the patterns of TiH₂ phase in its XRD patterns (Figure S4b, Supporting Information). Since the formation of titanium hydrides (e.g., TiH₂, TiH_{1.971}) reduces the mechanical properties of a Ti substrate,^[19]

XRD measurements were then applied to check the presence of titanium hydrides on the backside of a porous TiC/BDD composite film. In its XRD spectrum (Figure S4a, Supporting Information), no patterns for titanium hydrides were found. This is reasonable, due to the following experimental facts: i) the TiO₂ layer was formed in both sides of a Ti substrate since the Ti plate was vertically inserted in the electrolyte and its both sides were fully exposed to the electrolytes during the PEO process; ii) there was no or low exposure possibility of this substrate to hydrogen during CVD growth process since it laid down horizontally on the sample stage.

Note that the XRD patterns of a TiC film (Figure S4c, Supporting Information) reveal the presence of titanium oxides (e.g., Ti₂O₃) but their diffraction intensity is relatively weak. Consequently, the substitution of oxygen atoms inside the TiO₂ matrix with carbon atoms might be the key reaction during the CVD growth process. Fortunately, no diffraction signals of titanium oxides (e.g., Ti₂O₃) are found in the XRD patterns of a porous TiC/BDD composite film (Figure 2A-b). In other words, during the course of CVD growth of BDD, TiO₂ is fully converted into TiC and there is no titanium oxides (e.g., Ti₂O₃) left. Such a full conversion is because diamond nanoparticles seeded on a PEO-treated Ti substrate provide more carbon centers or reaction sites, accelerate the substitution efficiency/rate of oxygen inside the TiO₂ matrix with carbon during CVD growth of BDD, eventually leading to the formation of a composite film consisting of only TiC and BDD phases.

Since Raman spectroscopy is more sensitive to diamond and graphite in comparison to XRD, the Raman spectra of a TiC film and a porous TiC/BDD composite film were recorded (Figure 2B). For both films, two characteristic carbon peaks at ≈ 1336 and $\approx 1608\text{ cm}^{-1}$ appear, corresponding to the D and G bands, respectively. The G band is invented from the in-plane vibration of sp^2 carbon atoms, an indication of graphitic carbons in the TiC layer. The D band is related to sp^3 carbon bonding in BDD. As expected, the intensity ratio of D band to G band, obtained by Lorentzian fitting of the Raman spectrum of a TiC film (Figure 2B-a), is lower than that of a porous TiC/BDD composite film (Figure 2B-b). This is obviously due to the presence of BDD in the porous TiC/BDD composite film.

It has to point out that the intensities of characteristic XRD patterns of diamond are much lower, in comparison to those of a highly crystallized TiC film. A strong Raman signal of TiC is thus expected to be shown below the wavelength of 500 nm .^[20] Surprisingly, such a Raman signal has not been noticed for a porous TiC/BDD composite film, probably due to varied quality of TiC in the composite film^[20] and changed internal stress of this composite film.^[21] Namely, low crystallized and porous TiC phase in the TiC/BDD composite film might result in weak Raman signal, similarly as these reported for TiC powders.^[20] The high internal stress of this porous TiC/BDD composite film (more than hundreds of MPa), originating from the rapid solidification of molten oxide in the relatively cool electrolyte,^[22] reduces further Raman signal of the TiC phase and makes it becomes fully invisible.

The chemical composition and surface chemical states of a porous TiC/BDD composite film were further characterized using X-ray photoelectron spectroscopy (XPS). Figure 2C presents its high-resolution XPS spectra for C 1s (left) and Ti 2p (right). The C 1s signal is deconvoluted into four contributions, shown as C–Ti, C–C (sp^2), C–C (sp^3), and C–O with the binding energy of 281.9, 284.0, 285.1, and 286.6 eV, respectively.^[22] The sp^2 and sp^3 carbons again confirm the existence of amorphous carbon phase (namely TiC) and diamond phase (namely BDD) in a porous TiC/BDD composite film, respectively. The XPS spectrum of Ti 2p shows two main peaks located at 454.9 ($2\text{p}_{3/2}$) and 460.7 ($2\text{p}_{1/2}$) eV. Both are corresponding to a TiC phase.^[23] The shoulder located at 457.8 eV in between two main peaks indicates the existence of Ti ions with lower valences. These results are well consistent with those obtained from XRD and Raman measurements, further confirming the successful growth of a porous TiC/BDD composite film using the proposed approach.

Based on above results, the growth mechanism of a porous TiC/BDD composite film is discussed as following. In the first step, a TiO_2 layer is formed on the Ti substrate by means of PEO treatment. Such a TiO_2 layer is only nanometer-thick, porous, and possesses various micropores. It thus facilitates efficiently the seeding of diamond nanoparticles and eventually the CVD deposition of BDD. At least three advantages are expected from such a porous TiO_2 layer for the formation of a TiC/BDD composite film: i) this porous structure increases enormously microroughness, boosting the adhesion possibility of diamond nanoparticle seeds on the substrate; ii) the presence of oxides on such a film enhances diamond nucleation,

leading to enhanced density of diamond nanoparticle seeds on the substrate; iii) the CVD growth of BDD on such a porous layer can occur at a reduced growth temperature.^[24] In the second step, the porous TiO_2 layer is slightly etched/reduced by hydrogen plasma at an initial stage of this CVD process, bringing in more dips and increased surface roughness. After methane is introduced into the CVD chamber, the transformation of TiO_2 to TiC starts immediately.^[11] This is because methane and hydrogen are dissociated in the plasma, forming highly active hydrocarbon ions and radicals. Moreover, part of carbon atoms from diamond nanoparticles also dissolves into the TiO_2 matrix. These processes make full conversion of TiO_2 into TiC possible, rapid, and efficient. Note that the porous structure formed during the PEO process also facilitates the diffusion of hydrocarbons and radicals, leading to accelerated carbide formation.^[25] In comparison to TiO_2 , TiC is known to be more resistive toward hydrogen plasma etching and thus it is more stable under hydrogen atmosphere. Once hydrogen etching of TiC is suppressed, more hydrocarbon ions and radicals can diffuse through TiO_2 pores, leading to conversion of rest TiO_2 into TiC. On the other hand, the concentrations of generated hydrocarbon ions and radicals during CVD process are high. Consequently, a large amount of superficial carbon active species can go into the porous TiO_2 layer, resulting in highly efficient and fast conversion of TiO_2 into TiC. Once the porous TiO_2 layer is fully converted into a porous TiC layer, the TiC surface is immediately saturated with carbon species. The growth of BDD nanoparticles/nanocrystals then starts and finally dominates the whole process.^[26] Depending on the applied growth time, a porous or packed TiC/BDD composite film is formed. This is the third step. Since the formation of TiC is actually an equilibrium state between hydrogen plasma etching of TiC and diffusion of carbon atoms into TiO_2 , there are no clear boarders between these reactions described in the second and third steps.

2.2. Supercapacitor Performance of Porous TiC/BDD Composite Films

2.2.1. Capacitances

To demonstrate the potential applications of a porous TiC/BDD composite film for the construction of high-performance SCs, its capacitive behavior (Figure 3) was investigated using a three-electrode system in both aqueous (here $1\text{ M Na}_2\text{SO}_4$) and organic (here 0.1 M TBABAF_4 in acetonitrile, ACN) solutions. Both cyclic voltammetry (Figure 3a,c) and galvanostatic charge/discharge (GCD) method (Figure 3b,d) were applied. The cyclic voltammograms (CVs) recorded in two media at various scan rates (Figure 3a,c) exhibit nearly rectangular shapes, indicating ideal EDLC behavior of a porous TiC/BDD composite film under those conditions. Compared to an aqueous solution, an organic one provides a wider potential window but usually owns a lower conductivity in most cases,^[27] which might hinder the charge propagation on the porous TiC/BDD composite film. Consequently, a slight distortion is observed for the CVs recorded in the organic solutions. The CVs of a TiC electrode (Figure S5a–d, Supporting Information) and a BDD electrode

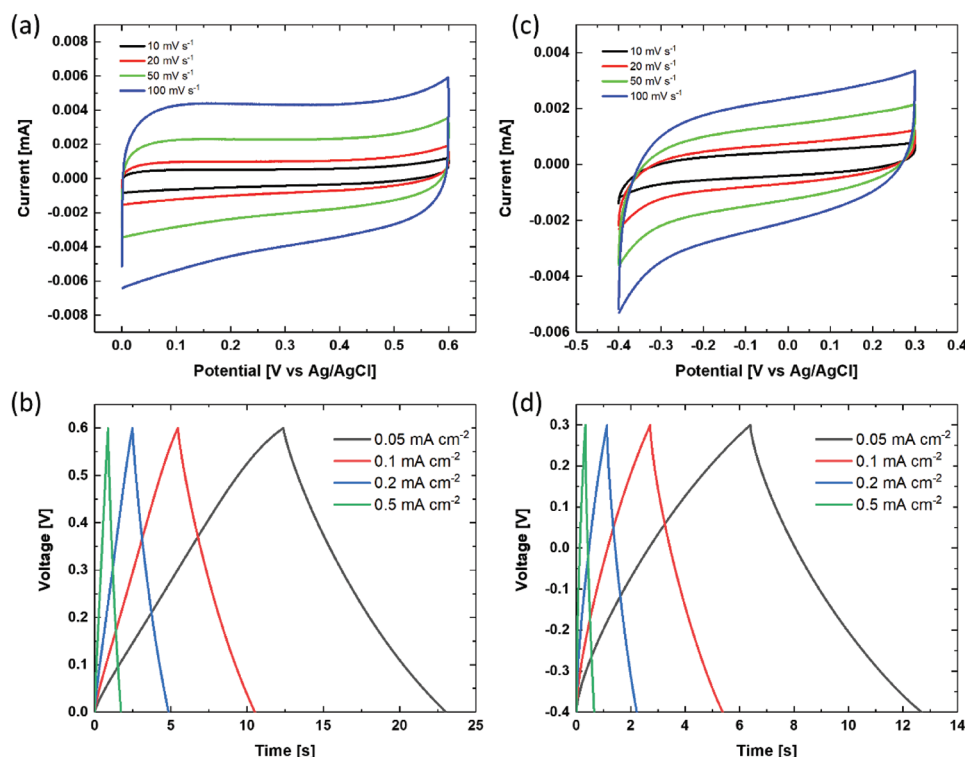


Figure 3. Performance of the porous TiC/BDD composite film based EDLCs in a,b) 1.0 M Na₂SO₄ and c,d) 0.1 M TBABAF₄/ACN: a,c) CVs at different scan rates and b,d) GCD curves at different current densities.

(Figure S6a–d, Supporting Information) also reveal good EDLC behavior. Their capacitances are much smaller than those of a porous TiC/BDD composite electrode at the same scan rates. In 1 M Na₂SO₄, the calculated capacitances of a porous TiC/BDD composite electrode are 1.0, 0.9, 0.8, and 0.7 mF cm⁻² at the scan rate of 10, 20, 50, and 100 mV s⁻¹, respectively. In 0.1 M TBABAF₄/ACN, they change to 0.8, 0.7, 0.5, 0.4 mF cm⁻², respectively.

Figure 3b,d show the GCD curves of a porous TiC/BDD composite film at different current densities in the aqueous and organic media, respectively. All GCD curves show fair triangles in shape, indicating high reversibility of these EDLCs. The IR drops can barely be observed at all applied current densities, resulting from the excellent conductivity of this composite film. At the current density of 0.05, 0.1, 0.2, and 0.5 mA cm⁻², the estimated capacitances in 1 M Na₂SO₄ are 1.0, 0.9, 0.8, and 0.7 mF cm⁻², respectively. Under the same current densities, they become 0.5, 0.4, 0.3, and 0.2 mF cm⁻² in 0.1 M TBABAF₄/ACN, respectively. Note that, under the same charging/discharge conditions, i) the capacitance of a porous TiC/BDD composite film in this aqueous solution is bigger than that in the used organic solution; ii) the capacitance of a porous TiC/BDD composite film is ≈2 times and even ≈2 orders of magnitude higher than that of a TiC electrode (Figure S5a–d, Supporting Information) and a BDD electrode (Figure S6a–d, Supporting Information), respectively. These results indicate the thicknesses and amounts of accumulated ions are varied on different capacitor electrodes and in different media. The porous TiC/BDD composite film

exhibits significantly the best EDLC performance among these electrodes.

To further improve the capacitive performance of a porous TiC/BDD composite film (e.g., capacitance, etc.), the construction of PCs by use of redox electrolytes was performed.^[6,7] Soluble redox probes such as 0.05 M Fe(CN)₆^{3-/4-} and 20 × 10⁻³ M ferrocene were added into 1 M Na₂SO₄ aqueous and 0.1 M TBABAF₄/ACN organic solutions, respectively. The CVs in Figure 4a,b show a pair of remarkable peaks at all scan rates, corresponding to Faradaic reactions of two redox species on this porous TiC/BDD composite film. The absolute values of the oxidation peak currents are almost identical to those of the reduction ones. The redox peak potentials of Fe(CN)₆^{3-/4-} redox couple are located at about 0.35 and 0.24 V, while those of ferrocene appear at about 0.57 and 0.29 V. The separation of two redox peak potentials (ΔE_p) in the aqueous solution is narrower than that in the organic one, indicating better reversibility of investigated redox reactions in an aqueous solution. This is due to varied electron transfer rates of two redox reactions as well as altered interactions between redox probes and the porous TiC/BDD composite film. On this porous composite film, the electrode kinetics of multiple-charged Fe(CN)₆^{3-/4-} inorganic ions in the aqueous solution seems to be faster than that of organic ferrocene molecule in the organic solution. The estimated capacitances in the aqueous solution are 42.6, 35.3, 24.6, and 18.0 mF cm⁻² at the scan rate of 10, 20, 50, and 100 mV s⁻¹, respectively. In the organic solution, they change to 21.4, 16.5, 10.9, and 7.8 mF cm⁻², respectively.

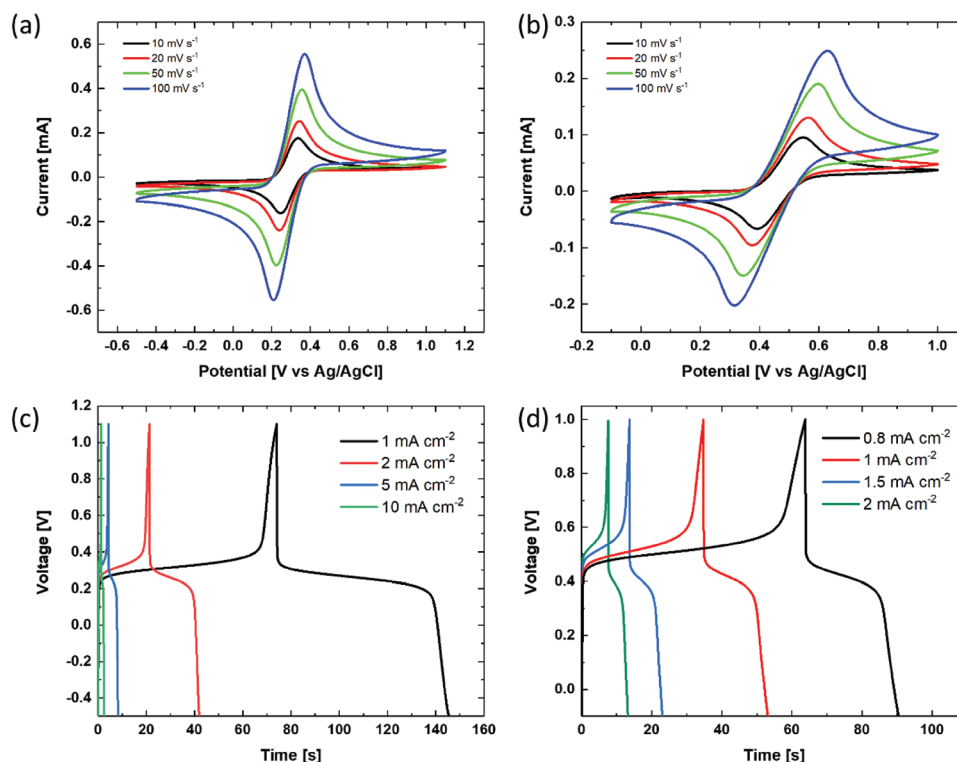


Figure 4. Performance of the porous TiC/BDD composite film based PCs in a,c) 0.05 M $\text{Fe}(\text{CN})_6^{3-/4-}$ + 1 M Na_2SO_4 aqueous solution and b,d) 20×10^{-3} M ferrocene + 0.1 M $\text{TBABF}_4/\text{ACN}$: a,b) CVs at different scan rates and c,d) GCD curves at different current densities.

The related GCD curves (Figure 4c,d) show nonlinear shapes. The average capacitances calculated by integrating GCD curves are 46.3, 26.7, 13.1, and 7.8 mF cm^{-2} in the aqueous solution at the current density of 1.0, 2.0, 5.0, and 10 mA cm^{-2} , respectively. They change to 22.7, 10.9, 9.2, and 5.8 mF cm^{-2} in the organic solution at the current density of 0.8, 1.0, 1.5, and 2 mA cm^{-2} , respectively. The CVs and GCD curves of a TiC electrode (Figure S5e–h, Supporting Information) and a BDD electrode (Figure S6e–h, Supporting Information) were also measured under the identified conditions. The estimated capacitances of a TiC electrode in redox species contained aqueous solution are 40.3, 33.0, 22.8, and 16.4 mF cm^{-2} at the scan rate of 10, 20, 50, and 100 mV s^{-1} , respectively. For a BDD electrode, they are 38.0, 29.0, 18.8, and 12.6 mF cm^{-2} , respectively.

To further explain the difference of these capacitances, the Nyquist plots of a porous TiC/BDD composite electrode, a TiC electrode, and a BDD electrode were recorded in 0.05 M $\text{Fe}(\text{CN})_6^{3-/4-}$ + 1 M Na_2SO_4 aqueous solution at open circuit potentials (Figure S7, Supporting Information). The variation of the charge-transfer resistances of three electrodes, estimated from the diameters of the semicircles in their Nyquist plots, follows the same variation tendency of their capacitances. Compared to those obtained in the solutions without adding redox electrolytes (namely the capacitances of EDLCs in Figure 3), the capacitances in the redox species contained solutions (namely, the pseudocapacitances of PCs in Figure 4) are dramatically enhanced. This is quite reasonable since fast electron transfer reactions of redox species occur at the electrode/electrolyte

interface and contribute dramatically into the obtained capacitances, leading to the formation of battery-like SCs.^[6–8]

2.2.2. Cyclic Stability

As an important characteristic for practical applications of SCs, the cyclic stability of the porous TiC/BDD composite film based SCs was tested by means of the GCD technique in both aqueous redox and organic solutions. The cyclic stability of PCs was investigated as a case study where a cycle number of 10 000 and a current density of 20 mA cm^{-2} were applied. The capacitance obtained in the aqueous solution shows dramatically better cyclic stability than that in the organic medium. In the aqueous solution, the capacitance of a porous TiC/BDD composite film based PC shows only about 8% loss of the initial capacitance after 10 000 cycles (Figure 5a), while a TiC film based PC loses about 18% of the initial capacitance (Figure S8, Supporting Information). This confirms better cyclic stability of a porous TiC/BDD composite film based SC, which is not attainable by conventional TiC based EES devices.^[13] This is because TiC is possible to be oxidized,^[28] resulting in reduced conductivity and eventually reduced capacitance retention for the TiC film based SCs. On the other hand, it has been shown that a higher TiC content is better for better capacitance retention of a TiC film based SCs.^[29] For a porous TiC/BDD composite film, the oxidation of TiC is partially prevented since the superficial BDD crystallites on the TiC layer form an isolated structure and thus direct contact of TiC with the electrolyte can

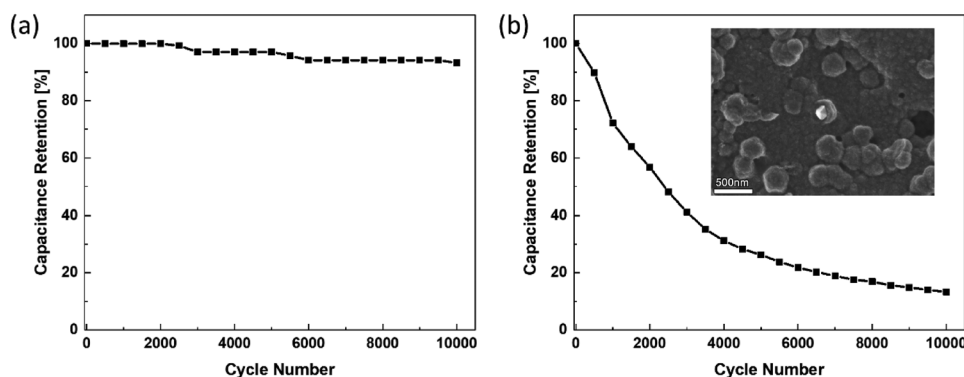


Figure 5. Capacitance retention of the porous TiC/BDD composite film based SCs at the current density of 20 mA cm^{-2} in a) $0.05 \text{ M Fe(CN)}_6^{3-/4-} + 1 \text{ M Na}_2\text{SO}_4$ aqueous solution and b) $20 \times 10^{-3} \text{ M ferrocene} + 0.1 \text{ M TBABF}_4/\text{ACN}$. The inset in (b) is the SEM image of a porous TiC/BDD composite film after 10 000 charge/discharge cycles.

be avoided efficiently. In other words, the oxidation of TiC in a porous TiC/BDD composite film is hindered and minimized. In conclusion, the porous TiC/BDD composite electrode has high stability, leading to its excellent capacitance retention.

Surprisingly, only about 16% capacitance remains after 10 000 cycles for the porous TiC/BDD film based PCs when the redox species contained organic solution is applied (Figure 5b). The surface of the porous TiC/BDD composite film was then examined using SEM (the inset of Figure 5b) and XPS (Figure S9, Supporting Information). Different from that shown in Figure 1c, its surface is partially coated with a layer (the inset of Figure 5b) after such a cyclic stability test. The XPS spectrum of this layer exhibits clearly a Fe-element related XPS peak, which is not seen on a porous TiC/BDD composite film prior to this cycle stability test (Figure S9, Supporting Information). Such a layer is probably an adsorbed or polymerized layer from unstable redox species, leading to hindered electron transfer processes and poor capacitance retention of constructed PCs in this redox species contained organic solution.

2.2.3. Energy and Power Densities

The energy density and power densities of this porous TiC/BDD composite film based SCs were estimated using a two-electrode system. A symmetrical SC device was constructed according to the method reported.^[30] The GCD curves were recorded at different current densities only in the aqueous solution. The achieved energy and power densities are summarized in Figure 6 as a Ragone plot. Clearly, the porous TiC/BDD composite film based PCs have higher energy and power densities than the porous TiC/BDD composite film based EDLCs. For a PC, the maximum energy density obtained in the aqueous solution is about 47.4 Wh kg^{-1} together with a power density of 2236 W kg^{-1} . The energy density remains 5.7 Wh kg^{-1} even at a high power density of $21\,918 \text{ W kg}^{-1}$. Namely, this ultra-high energy density is reached without sacrifice of its power density. Compared to other available EES devices (e.g., EDLCs, batteries, etc.), this porous TiC/BDD composite film based PC exhibits not only much higher power density but also much higher energy density. The obtained energy density is even higher than that of some lithium-ion batteries.

It is actually not surprising that this porous TiC/BDD composite film exhibits such excellent capacitive performance. There are several facts behind. First, such a capacitor electrode is free of organic binders and current collectors. Second, it is ultrathin (only about 750 nm). In other words, the mass of the capacitor electrode is significantly reduced. Third, this capacitor electrode owns a broad pore-size distribution (ranging from nanopores to micropores), beneficial for charge accumulation, electron transfer, and mass transport. More importantly, these pores can be perfect pockets to encapsulate redox electrolytes. The pores with large sizes provide pathways for ion transport and those with small dimensions are useful for ion reservoirs. The minimized diffusion distance to the interior surfaces is thus expected, accelerating efficiently the kinetic process of the ion diffusion and mass transport. Finally, the TiC phase in the

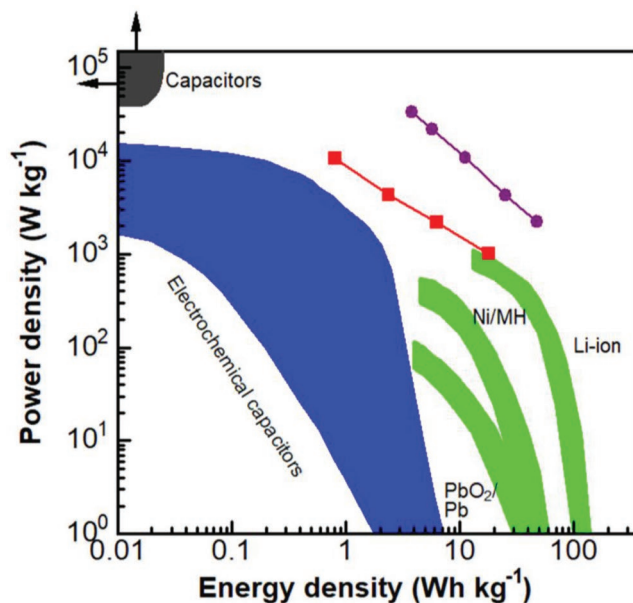


Figure 6. Comparison of Ragone plots of the porous TiC/BDD composite film based EDLCs (red line) as well as PCs (violet line) in the aqueous solution with that of traditional capacitors, available electrochemical capacitors, and different batteries. Reproduced with permission.^[31] Copyright 2008, Nature Publisher.

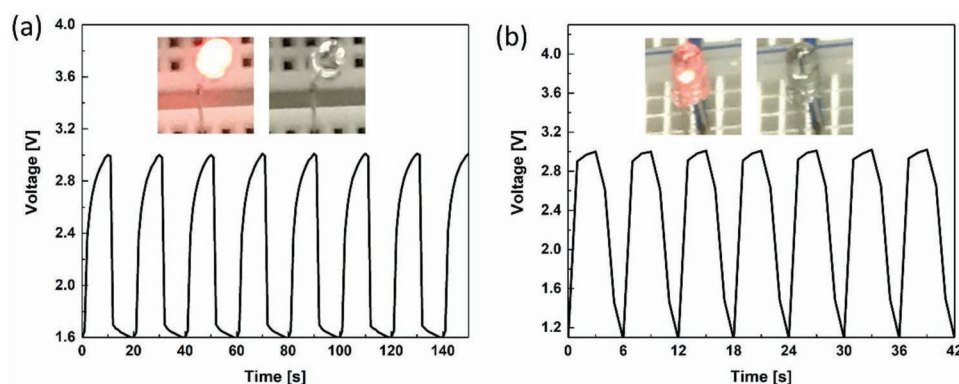


Figure 7. Voltage–time curves during the course of charging/discharging processes for a) an EDLC demonstrator and b) a PC demonstrator. The insets are the photos of the red LED during the course of discharging (left) and charging (right) processes.

composite film facilitates the electrical conductivity of this composite film as well as contributes the rapid ion and electron transportation.

2.3. Supercapacitor Demonstrator

SC demonstrators were fabricated using either porous TiC/BDD composite film based EDLCs or porous TiC/BDD composite film based PCs. **Figure 7a,b** show the voltage–time curves recorded on an EDLC demonstrate and a PC one, respectively. Excellent reversibility with rapid charging and discharging rates is found for both demonstrators. For the EDLC demonstrator, the red LED lighted up continuously for 10 s after a charging time of 12 s. For PC, the red LED lighted for 3 s with a charging time of 3 s. These supercapacitor devices demonstrate the potential of applying the porous TiC/BDD composite film as the capacitor electrode to construct high-performance EDLC and PC devices for practical applications.

3. Conclusion

A porous TiC/BDD composite film has been utilized as the capacitor electrode to construct high-performance supercapacitors. Chemical vapor deposition of boron-doped diamond on a porous and nanometer-thick titanium oxide layer formed during a plasma electrolytic oxidation leads to the generation of this porous TiC/BDD composite film. Such a technique allows fabricating capacitor electrodes without the use of binders or extra current collectors that are often required for traditional sp^2 carbon materials. Meanwhile, this technique successfully overcomes the drawback of low capacitance retention of TiC capacitor electrodes. Together with soluble redox electrolytes, battery-like supercapacitors have been constructed, featuring big and stable capacitances, high power densities, rapid charging/discharging rates, easy assembling ability, and especially extremely high energy densities. Therefore, this porous TiC/BDD composite film will be highly useful in the design of next-generation supercapacitor systems in both aqueous and organic solutions. Although further activities can be conducted

such as the study of the encapsulation behavior of redox electrolytes inside the pores, this strategy offers a way to design and synthesize new capacitor electrodes for the construction of high-performance and battery-like supercapacitors, supercapacitor batteries, or supercapacitor batteries.

4. Experimental Section

Growth and Characterization of Composite Films: A two-step approach was developed to synthesize porous TiC/BDD composite films. The first step was the PEO treatment of Ti substrates. After being cut into the suitable rectangular dimensions (e.g., $1 \times 2 \text{ cm}^2$), the highly pure Ti plates (99.6%) with a thickness of 2 mm (Alfa Aesar, Germany) were ground and polished with a 1200#-SiC abrasive paper. It was then washed and cleaned successively with distilled water, acetone, and Turco 5578. Subsequently, a PEO process was applied using a DC power supply (EA-PSI 8720, Conrad Electronics, Germany). A two-electrode system was applied where a Ti plate was the anode and a cylindrical stainless steel was the cathode. A constant voltage mode was employed to form TiO_2 layers on the Ti substrate. The used electrolyte, anodic power voltage, and anodic oxidation time was 0.066 M Na_3PO_4 aqueous solution, 280 V, and 15 min, respectively. During the oxidation process, the electrolyte was cooled to 10°C by use of a circulation cooling system. A current of 1 A was applied to secure the circuit. After that, the oxidized Ti substrate was immersed into a suspension of detonation nanodiamonds (New Metals & Chemicals Corporation, Japan) in an ultrasonic bath for 10 min. The detonation nanodiamonds with an average size of 5 nm served as diamond seeds for the CVD growth of BDD. The BDD films were deposited in an ASTeX-A5000 microwave plasma enhanced CVD reactor. To maintain the stable plasma, hydrogen gas with a flow rate of 400 sccm was introduced into the chamber at the initial stage. Once the gas pressure and microwave power reached 45 Torr and 1400 W, respectively, a mixture of methane and trimethylborane gases (with a volume ratio of 1:2) was purged. The growth time was varied from 15 to 90 min. To grow a porous TiC/BDD composite film, the optimized growth time of 30 min was used. The TiC film was prepared on a PEO-treated Ti substrate through exposing it to a chamber under the conditions for the CVD growth of BDD (here no seeding process was applied). The growth of BDD films on Si wafers was conducted according to the previous reports.^[3]

The morphologies of the synthesized films were characterized using field emission scanning electron microscopy (FESEM, Zeiss ultra55, Germany). Their elemental compositions and chemical states were investigated with an XPS (Surface Science Instruments, SSX-100 S-probe photoelectron spectrometer, USA) using Al $K\alpha$ radiation of

200 W. A spot size of 800 μm^2 and a resolution of 1 eV were used for the survey scan between 0 and 1200 eV. For the high-resolution scans, the spot size was set to 300 μm^2 and the resolution to 0.1 eV. The XPS spectra were analyzed using CasaXPS processing software (version 2.317 PR 1.1). The structure and phase distribution of the synthesized films were analyzed by XRD (Philips X'Pert, Germany) using Cu K α radiation (40 kV, 40 mA).

Electrochemical Measurements: Electrochemical measurements were carried out on a CHI660E electrochemical workstation (Shanghai Chenhua Inc., China). In a three-electrode system, an Ag/AgCl electrode was used as the reference electrode and a Pt wire was as the counter electrode. The geometric area of used capacitor electrode was 0.05 cm^2 . Cyclic voltammograms (CV), galvanostatic charge/discharge (GCD) curves of different capacitor electrodes were recorded. To investigate the performance of aqueous and organic EDLCs, the solutions of 1.0 M Na_2SO_4 aqueous solution and 0.1 M tetrabutylammonium tetrafluoroborate (TBABF $_4$) in acetonitrile (ACN) were used, respectively. The performance of aqueous and organic PCs was investigated in 0.05 M $\text{K}_3\text{Fe}(\text{CN})_6/\text{K}_4\text{Fe}(\text{CN})_6$ + 1 M Na_2SO_4 and 20×10^{-3} M ferrocene + 0.1 M TBABF $_4$ /ACN, respectively.

To estimate the energy and power densities of the formed SCs, a two-electrode symmetrical SC device was constructed. A Nafion membrane with a thickness of 50 μm (Alfa Aesar, Germany) was used as the separator. The mass of the capacitor electrode was the sum of TiC and BDD weights, calculated using the equation of [density] $\times$ [thickness] $\times$ [exposed area] $\times$ [exposed ratio]. For TiC, its density, thickness, exposed area (namely electrode area), and expose ratio are 4.93 g cm^{-3} , 750 nm, 0.05 cm^2 , and 94% (=100% – 6%, due to the estimated porosity of 6%), respectively. For diamond, they are 3.75 g cm^{-3} , 200 nm, 0.05 cm^2 , and 75% (the estimated BDD coverage), respectively. The capacitance (C, F cm^{-2}), energy density (E, Wh kg^{-1}), and power density (P, W kg^{-1}) were calculated according to the reported methods.^[7] The equations for linear charge/discharge GCD curves (or rectangular CVs) and for nonlinear charge/discharge GCD curves (or peak-shaped CVs) were utilized to calculate these values of constructed EDLC and PC devices, respectively.

SC Demonstrators: The porous TiC/BDD composite film based cells were assembled in a series as a SC demonstrator to light up a red LED, of which working voltage is 1.8 V. The set up was similar as the one previously reported.^[30] Briefly, two TiC/BDD composite electrodes were pasted on the both sides of a cell that was separated in the middle of the cell by a 50 μm thick Nafion membrane (Alfa Aesar, Germany). The effective electrode area was 0.785 cm^2 . The solutions of 1 M Na_2SO_4 and 0.05 M $\text{K}_3\text{Fe}(\text{CN})_6/\text{K}_4\text{Fe}(\text{CN})_6$ + 1 M Na_2SO_4 were used to construct EDLCs and PCs, respectively. A SC demonstrator consisting of four porous TiC/BDD composite film based EDLCs or three porous TiC/BDD composite film PCs that were connected in a series. A single-board microcontroller (Arduino UNO) was employed to control charging/discharging processes of these demonstrators and to record the related voltage–time curves. By connecting this microcontroller with a USB cable to the computer, the charging/discharging processes of such demonstrators were programmed and automatically switched. For example, the demonstrator is automatically charged when the voltage is lower than 1.6 V. When the voltage is higher than 1.8 V, the red LED lights up.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors acknowledge the financial support from the German Research Foundation (DFG) under the projects of JI 22/25-1 (J.X.) and YA344/1-1 (N.Y.).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

boron-doped diamond, composite films, energy densities, high-performance supercapacitors, TiC

Received: November 22, 2018

Revised: February 4, 2019

Published online:

- [1] S. Ould Amrouche, D. Rekioua, T. Rekioua, S. Bacha, *Int. J. Hydrogen Energy* **2016**, *41*, 20914.
- [2] Y. Zhu, S. Murali, M. D. Stoller, K. J. Ganesh, W. Cai, P. J. Ferreira, A. Pirkle, R. M. Wallace, K. A. Cytochsz, M. Thommes, D. Su, E. A. Stach, R. S. Ruoff, *Science* **2011**, *332*, 1537.
- [3] E. Almeida, M. Baldan, J. Rosolen, N. Ferreira, *Diamond Relat. Mater.* **2008**, *17*, 1529.
- [4] J. Jang, J. Bae, M. Choi, S.-H. Yoon, *Carbon* **2005**, *43*, 2730.
- [5] D. N. Futaba, K. Hata, T. Yamada, T. Hiraoka, Y. Hayamizu, Y. Kakudate, O. Tanaike, H. Hatori, M. Yumura, S. Iijima, *Nat. Mater.* **2006**, *5*, 987.
- [6] a) N. Yang, S. Yu, J. V. Macpherson, Y. Einaga, H. Zhao, G. Zhao, G. M. Swain, X. Jiang, *Chem. Soc. Rev.* **2019**, *48*, 157; b) S. Yu, N. Yang, H. Zhuang, S. Mandal, O. A. Williams, B. Yang, N. Huang, X. Jiang, *J. Mater. Chem. A* **2017**, *5*, 1778; c) S. Yu, N. Yang, H. Zhuang, J. Meyer, S. Mandal, O. A. Williams, I. Lilje, H. Schönherr, X. Jiang, *J. Phys. Chem. C* **2015**, *119*, 18918; d) H. Zhuang, N. Yang, H. Fu, L. Zhang, C. Wang, N. Huang, X. Jiang, *ACS Appl. Mater. Interfaces* **2015**, *7*, 5384.
- [7] a) Y. Shao, M. F. El-Kady, J. Su, Y. Li, Q. Zhang, M. Zhu, H. Wang, B. Dunn, R. B. Kaner, *Chem. Rev.* **2018**, *118*, 9223; b) J. Lee, S. Choudhury, D. Wingarth, V. Presser, *ACS Appl. Mater. Interfaces* **2016**, *8*, 23676.
- [8] L. Sun, J. Gong, D. Zhu, Z. Zhu, S. He, *Adv. Mater.* **2004**, *16*, 1849.
- [9] K. Honda, T. N. Rao, D. Tryk, A. Fujishima, M. Watanabe, K. Yasui, H. Masuda, *J. Electrochem. Soc.* **2001**, *148*, A668.
- [10] Z. F. Zhang, X. P. Jia, S. S. Sun, X. B. Liu, Y. Li, B. M. Yan, H. A. Ma, *Int. J. Refract. Met. Hard Mater.* **2013**, *38*, 111.
- [11] a) K. Siuzdak, R. Bogdanowicz, M. Sawczk, M. Sobaszek, *Nanoscale* **2015**, *7*, 551; b) M. Sobaszek, K. Siuzdak, M. Sawczk, J. Ryl, R. Bogdanowicz, *Thin Solid Films* **2016**, *601*, 35; c) M. Sawczak, M. Sobaszek, K. Siudak, J. Ryl, R. Bogdanowicz, K. Darowicki, M. Gazda, A. Cenian, *J. Electrochem. Soc.* **2015**, *162*, A2085.
- [12] H. Zanin, P. W. May, D. Fermin, D. Plana, S. M. C. Vieira, W. I. Milne, E. J. Corat, *ACS Appl. Mater. Interfaces* **2014**, *6*, 990.
- [13] a) X. Wang, S. Kajiyama, H. Iinuma, E. Hosono, S. Oro, I. Moriguchi, M. Okubo, A. Yamada, *Nat. Commun.* **2015**, *6*, 6544; b) Y. Xie, Y. Dall'Agnese, M. Naguib, Y. Gogotsi, M. W. Barsoum, H. L. Zhuang, P. R. C. Kent, *ACS Nano* **2014**, *8*, 9606; c) W. Zhang, L. Zu, B. Kong, B. Chen, H. He, K. Lan, Y. Liu, J. Yang, D. Zhao, *iScience* **2018**, *3*, 149; d) Y. Yao, K. Huo, L. Hu, N. Liu, J. J. Cha, M. T. McDowell, P. K. Chu, Y. Cui, *ACS Nano* **2011**, *5*, 8346; e) X. Liang, A. Garsuch, F. Nazar Linda, *Angew. Chem., Int. Ed.* **2015**, *54*, 3907; f) J. Chmiola, C. Largeot, P.-L. Taberna, P. Simon, Y. Gogotsi, *Science* **2010**, *328*, 480.
- [14] a) S. Kondrat, C. R. Perez, V. Presser, Y. Gogotsi, A. A. Kornyshev, *Energy Environ. Sci.* **2012**, *5*, 6474; b) J. Chmiola, G. Yushin, R. Dash, Y. Gogotsi, *J. Power Sources* **2006**, *158*, 765.

- [15] Q. Li, J. Liang, Q. Wang, in *Modern Surface Engineering Treatments* (Ed: M. Aliofkhazraei), London, UK **2013**, pp. 75–99.
- [16] B. Yan, N. L. Loh, Y. Fu, C. Q. Sun, P. Hing, *Surf. Coat. Technol.* **1999**, 115, 256.
- [17] P. Huang, F. Wang, K. Xu, Y. Han, *Surf. Coat. Technol.* **2007**, 201, 5168.
- [18] a) D. Reyes-Coronado, G. Rodríguez-Gattorno, M. E. Espinosa-Pesqueira, C. Cab, R. d. Coss, G. Oskam, *Nanotechnology* **2008**, 19, 145605; b) Q. Zhang, L. Gao, J. Guo, *Appl. Catal., B* **2000**, 26, 207.
- [19] a) J. M. Chappé, A. C. Fernandes, C. Moura, E. Alves, N. P. Barradas, N. Martin, J. P. Espinós, F. Vaz, *Surf. Coat. Technol.* **2012**, 206, 2525; b) S. Ferro, M. Dal Colle, A. De Battisti, *Carbon* **2005**, 43, 1191; c) P. Mérel, M. Tabbal, M. Chaker, S. Moisa, J. Margot, *Appl. Surf. Sci.* **1998**, 136, 105.
- [20] a) B. H. Lohse, A. Calka, D. Wexler, *J. Appl. Phys.* **2005**, 97, 114912; b) B. Lohse, A. Calka, D. Wexler, *J. Alloys Compd.* **2007**, 434–435, 405.
- [21] a) J. W. Ager, M. D. Drory, *Phys. Rev. B* **1993**, 48, 2601; b) B. Dietrich, E. Bugiel, J. Klatt, G. Lippert, T. Morgenstern, H. J. Osten, P. Zaumseil, *J. Appl. Phys.* **1993**, 74, 3177.
- [22] a) Y. Guan, Y. Xia, F. Xu, *Surf. Coat. Technol.* **2008**, 202, 4204; b) R. H. U. Khan, A. L. Yerokhin, T. Pilkington, A. Leyland, A. Matthews, *Surf. Coat. Technol.* **2005**, 200, 1580.
- [23] X. Xia, Y. Zhang, D. Chao, Q. Xiong, Z. Fan, X. Tong, J. Tu, H. Zhang, H. J. Fan, *Energy Environ. Sci.* **2015**, 8, 1559.
- [24] S. T. Lee, Z. Lin, X. Jiang, *Mater. Sci. Eng., R* **1999**, 25, 123.
- [25] K. Kobashi, K. Nishimura, Y. Kawate, T. Horiuchi, *Phys. Rev. B* **1988**, 38, 4067.
- [26] H. Liu, D. S. Dandy, *Diamond Relat. Mater.* **1995**, 4, 1173.
- [27] C. Zhong, Y. Deng, W. Hu, J. Qiao, L. Zhang, J. Zhang, *Chem. Soc. Rev.* **2015**, 44, 7484.
- [28] K. Neuerer, *PhD. Thesis*, University of Erlangen-Nuremberg, **2013**.
- [29] E. Allcorn, A. Manthiram, *J. Mater. Chem. A* **2015**, 3, 3891.
- [30] S. Yu, N. Yang, M. Vogel, S. Mandal, O. A. Williams, S. Jiang, H. Schönherr, B. Yang, X. Jiang, *Adv. Energy Mater.* **2018**, 8, 1702947.
- [31] P. Simon, Y. Gogotsi, *Nat. Mater.* **2008**, 7, 845.